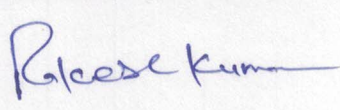
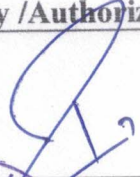




Test Report No: NEXT-20220810-07 IS 16047:2012/IEC61960:2011 Page 1 of 8
ULR: TC500122100001612F IS 16046(Part 2):2018/ Date: 21/09/2022
Group: Cells & Batteries IEC 62133-2:2017 Discipline: Electrical

Manufacturer Name and Address:	DAZZLE ROBOTICS PRIVATE LIMITED B1+B2+B3/5, GIDC ELECTRONICS ESTATE, GANDHINAGAR, SECTOR 25, K-5 CIRCLE, GANDHINAHAR, GUJARAT, India - 382044		
Product Name:	Rechargeable Li-Polymer Battery Pack		
Applicant Name and Address:	Garuda Aerospace Private Limited 24/46, K B Dasan road, Seetammal Colony, Seethammal Colony, Lubdhi Colony, Alwarpet, Chennai 600018		
Testing Laboratory Address:	NEXTRON INTERNATIONAL LAB PRIVATE LIMITED G-45, POCKET-C, SECTOR-4, BAWANA, NEW DELHI-110039		
Item No/ Model No:	22.2V 6S 16000mAh Tattu	Date of receipt:	10/08/2022
Country of origin:	India	Date of starting of test:	10/08/2022
Serial No:	Nil	Date of end of test:	21-09-2022
Job order No:	NEXT/20220810/07	Sample status:	OK/NOT OK

Applicable Standard	Test Description	Result
IS 16047:2012/IEC61960:2011	Discharge performance at 20°C (rated capacity)	Pass
IS 16046(Part 2):2018/ IEC 62133-2:2017	External Short Circuit, Over charge, Vibration & shock test	Pass

Tested by:	Approved by /Authorized Signature
	
RAKESH KUMAR (SR. TESTING ENGINEER) Date: 21/09/2022	PAWAN VERMA (TECHNICAL MANAGER) Date: 21/09/2022





Test Report No: NEXT-20220810-07

ULR: TC500122100001612F

Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

IS 16046(Part 2):2018/

IEC 62133-2:2017

Page 2 of 8

Date: 21/09/2022

Discipline: Electrical

GENERAL INFORMATION

1.Environment Condition:

Ambient Temperature	Ambient Humidity
(20±5) °C	(45 to 75) %RH

2.Product Rating:

Nominal voltage	22.2Vdc
Rated capacity	16000mAh
Charge cut-off voltage	25.2V
Discharge cut-off voltage	18.0V
Standard charging current	0.5C
Max. charging current	2C
Max. discharge current	112A
Weight	1.865 kg
Dimension of Battery Pack	Length: 190mm, Width: 75mm, Height: 63mm

R. K. Singh





NEXTRON INTERNATIONAL LAB PRIVATE LIMITED

NABL Accredited And BIS Recognized Test Lab

Regd. Office: G-45, POCKET-C, SECTOR-4, BAWANA NEW DELHI-110039

Web: www.nextronlab.com; E-Mail: Info@nextronlab.com; Mobile: +91 7678674846



Test Report No: NEXT-20220810-07
ULR: TC500122100001612F
Group: Cells & Batteries

IS 16047:2012/IEC61960:2011
IS 16046(Part 2):2018/
IEC 62133-2:2017

Page 3 of 8
Date: 21/09/2022
Discipline: Electrical

Copy of Marking Plate:

Rechargeable Li-Polymer Battery Pack

Model: 22.2V 6S 16000mAh Tattu

Voltage: 22.2V

Capacity: 16000mAh

6ICP63/75/190

DAZZLE ROBOTICS PRIVATE LIMITED

Mfg: 08/2022

Red (+)

Black (-)

Caution:

Use the authorized charger only.

Never disassemble by yourself.

Never short-circuit the battery.

Dispose of the battery properly.

Expose battery to open flames
could cause explosion.



MARKING PLATE OF BATTERY

Rose Kumar



1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing.



Test Report No: NEXT-20220810-07	IS 16047:2012/IEC61960:2011	Page 4 of 8
ULR: TC500122100001612F	IS 16046(Part 2):2018/	Date: 21/09/2022
Group: Cells & Batteries	IEC 62133-2:2017	Discipline: Electrical

TEST SPECIFICATIONS

1.	Discharge performance at 20°C (rated capacity)		
2.	Test purpose		
	This test verified the rated capacity of a cell or battery		
3.	Test procedure		
	Step 1- The cell or battery shall be charged in accordance with 7.2. Step 2- The cell or battery shall be stored, in an ambient temperature of 20°C±5 °C, for not less than 1 h and not more than 4 h. Step 3- The cell or battery shall be discharge, in an ambient temperature of 20°C±5 °C, at a constant current of 0,2 It A, until its voltage is equal to the specified end -of- discharge voltage.		
4.	Acceptance criteria		
	The capacity (Ah) delivered during step 3 shall be not less than 100% of the rated capacity declared by the manufacturer.		
5.	Result		
Number of test cycles performed (Step 1 to Step 3)		Capacity delivered after step 3	Rated capacity declared by the manufacturer
1		16.240 Ah	16Ah
			Test result
			Pass

Rakesh Kumar





Test Report No: NEXT-20220810-07
 ULR: TC500122100001612F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011
 IS 16046(Part 2):2018/
 IEC 62133-2:2017

Page 5 of 8
 Date: 21/09/2022
 Discipline: Electrical

TEST SPECIFICATIONS

Clause	Requirement + Test	Result - Remark	Verdict
7.3.2	External short-circuit (battery)	Complied	P
	The batteries were tested until one of the following occurred:	See below	P
	- 24 hours elapsed; or	Batteries were tested for 24h	P
	- The case temperature declined by 20 % of the maximum temperature rise	See above	N/A
	In case of rapid decline in short circuit current, the battery pack remained on test for an additional one hour after the current reached a low end steady state condition	No such condition occurred	N/A
	A single fault in the discharge protection circuit conducted on one to four (depending upon the protection circuit) of the five samples before conducting the short-circuit test		N/A
	A single fault applies to protective component parts such as MOSFET, fuse, thermostat or positive temperature coefficient (PTC) thermistor	See table 7.3.2	P
	Results: No fire. No explosion*.....:	No fire, No explosion during and after the test	P
7.3.6	Over-charging of battery	Complied	P
	The supply voltage which is:	See below	P
	- 1,4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6,0 V) for single cell/cell block batteries or	Series connected multi cell battery	N/A
	- 1,2 times the upper limit charging voltage resented in Table A.1 per cell for series connected multi-cell batteries, and	Complied	P
	- Sufficient to maintain a current of 2,0 It A throughout the duration of the test or until the supply voltage is reached	Complied	P
	Test was continued until the temperature of the outer casing:	See below	P
	- Reached steady state conditions (less than 10 °C change in 30-minute period); or	See below	N/A
	- Returned to ambient	Temperature of outer casing returned to ambient temperature	P
	Results: No fire. No explosion*.....:	No fire, no explosion during and after the test	P
7.3.8	Mechanical tests (batteries)	See below	P
7.3.8.1	Vibration	Complied	P
	Results: No fire, no explosion, no rupture, no leakage or venting*.	No fire, no explosion, no rupture, no leakage or venting during and after the test	P
7.3.8.2	Mechanical shock	Complied	P
	Results: No leakage, no venting, no rupture, no explosion and no fire *..... :	No fire, no explosion, no rupture, no leakage or venting during and after the test	P



1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing.



NEXTRON INTERNATIONAL LAB PRIVATE LIMITED

NABL Accredited And BIS Recognized Test Lab

Regd. Office: G-45, POCKET-C, SECTOR-4, BAWANA NEW DELHI-110039

Web: www.nextronlab.com; E-Mail: Info@nextronlab.com; Mobile: +91 7678674846



Test Report No: NEXT-20220810-07	IS 16047:2012/IEC61960:2011	Page 6 of 8
ULR: TC500122100001612F	IS 16046(Part 2):2018/	Date: 21/09/2022
Group: Cells & Batteries	IEC 62133-2:2017	Discipline: Electrical

7.3.2	TABLE: External short-circuit (battery)						P
Sample no.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maximum case temperature rise ΔT (°C)	Component single fault condition	Results	
22.2V 6S 16000mAh Tattu	22	25.2	80mΩ±20mΩ	0.6	NORMAL	A,B,C,D,E,F,G	
Supplementary information: - A: No fire or explosion - B: No leakage - C: No rupture - D: No venting - E: Bulge - F: Fire - G: Others (The test was completed after 24h)							

7.3.6	TABLE: Over-charging of battery				P
Constant charging current (A).....:				32.0	---
Supply voltage (Vdc).....:				36.0	---
Sample no.	OCV before charging (Vdc)	Total charging time (minute)	Maximum outer case temperature (°C)	Results	
22.2V 6S 16000mAh Tattu	18.1	--	27.1	A	
Supplementary information: Nil - A: No fire or explosion - B: No leakage - C: No rupture - D: No venting - E: Bulge - F: Fire - G: Others (please explain)					

Prese kune



1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing.



NEXTRON INTERNATIONAL LAB PRIVATE LIMITED

NABL Accredited And BIS Recognized Test Lab

Regd. Office: G-45, POCKET-C, SECTOR-4, BAWANA NEW DELHI-110039

Web: www.nextronlab.com; E-Mail: Info@nextronlab.com; Mobile: +91 7678674846



Test Report No: NEXT-20220810-07	IS 16047:2012/IEC61960:2011	Page 7 of 8
ULR: TC500122100001612F	IS 16046(Part 2):2018/	Date: 21/09/2022
Group: Cells & Batteries	IEC 62133-2:2017	Discipline: Electrical

7.3.8.1	TABLE: Vibration					P
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test (g)	Results	
22.2V 6S 16000mAh Tattu	25.2	25.2	1863	1863	A, B, C, D	
Supplementary information: Nil - A: No fire or explosion - B: No leakage - C: No rupture - D: No venting - E: Bulge - F: Fire - G: Others (please explain)						

7.3.8.2	TABLE: Mechanical shock					P
Sample no.	OCV before test (Vdc)	OCV after test (Vdc)	Mass before test (g)	Mass after test (g)	Results	
22.2V 6S 16000mAh Tattu	25.2	25.2	1863	1863	A, B, C, D	
Supplementary information: Nil - A: No fire or explosion - B: No leakage - C: No rupture - D: No venting - E: Bulge - F: Fire - G: Others (please explain)						

Prashant Kumar





Test Report No: NEXT-20220810-07
ULR: TC500122100001612F
Group: Cells & Batteries

IS 16047:2012/IEC61960:2011
IS 16046(Part 2):2018/
IEC 62133-2:2017

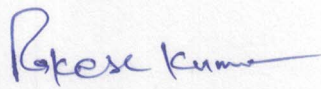
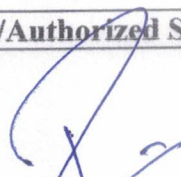
Page 8 of 8
Date: 21/09/2022
Discipline: Electrical

Attachment 1

Photo Documents:



BATTERY VIEW

Tested by:	Approved by /Authorized Signature
	
RAKESH KUMAR (SR. TESTING ENGINEER)	PAWAN VERMA (TECHNICAL MANAGER)
Date: 21/09/2022	Date: 21/09/2022

*****END OF TEST REPORT*****





TC-5696

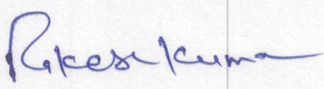
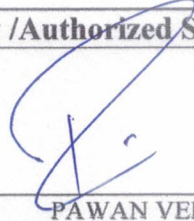
Test Report No: NEXT-20220713-04
ULR: TC569622100001686F
Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 1 of 8
Date: 08/11/2022
Discipline: Electrical

Manufacturer Name and Address:	DAZZLE ROBOTICS PRIVATE LIMITED B1+B2+B3/5, GIDC ELECTRONICS ESTATE, GANDHINAGAR, SECTOR 25, K-5 CIRCLE, GANDHINAGAR, GUJARAT, India – 382044		
Product Name:	Rechargeable Li-Polymer Battery Pack		
Applicant Name and Address:	Garuda Aerospace Private Limited 24/46, K B Dasan road, Seetammal Colony, Seethammal Colony, Lubdhi Colony, Alwarpet, Chennai 600018		
Testing Laboratory Address:	NEXTRON INTERNATIONAL LAB PRIVATE LIMITED G-45, POCKET-C, SECTOR-4, BAWANA, NEW DELHI-110039		
Item No/ Model No:	44.4V 12S 16000mAh Tattu	Date of receipt:	13/07/2022
Country of origin:	India	Date of starting of test:	13/07/2022
Serial No:	Nil	Date of end of test:	08/11/2022
Job order No:	NEXT/20220713/04	Sample status:	OK/NOT OK

Applicable Standard	Test Description	Result
IS 16047:2012/IEC61960:2011	Endurance in Cycle	Pass

Tested by:	Approved by /Authorized Signature
	
RAKESH KUMAR (SR. TESTING ENGINEER)	PAWAN VERMA (TECHNICAL MANAGER)
Date: 08/11/2022	Date: 08/11/2022



1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing



TC-5696

Test Report No: NEXT-20220713-04
ULR: TC569622100001686F
Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

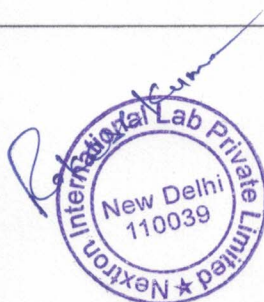
Page 2 of 8
Date: 08/11/2022
Discipline: Electrical

1.Environment Condition:

Ambient Temperature	Ambient Humidity
(20±5) °C	(45 to 75) %RH

2.Product Rating:

Nominal voltage	44.4Vdc
Rated capacity	16000mAh
Charge cut-off voltage	50.4V
Discharge cut-off voltage	44.4V
Standard charging current	0.5C
Max. charging current	2C
Max. discharge current	112A



1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing



TC-5696

Test Report No: NEXT-20220713-04
 ULR: TC569622100001686F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 3 of 8
 Date: 08/11/2022
 Discipline: Electrical

7.6	Endurance in cycles			
	Acceptance criteria			
	The capacity (Ah) measured according to step 1 to step 3 of 7.3.1 and not less than 80% C ₅ Ah of the rated capacity declared by the manufacturer.			
	Result			
Cycle number	Measured capacity after cycle	Charge	Stand in charge condition (h)	Discharge
250	13.120 Ah	(CC/CV) charging method declared by the manufacturer	50min	112A to final voltage (declared by the manufacturer)
Note: As 80% of 16Ah is 12.8Ah and on 250 cycle the DOD was noted to be 13.120Ah. Hence, as per the result the DOD is greater than 80%.				

Cycle Number	Charge cut-off voltage	Dis-Charge cut-off voltage	Discharge Rate	Initial Charge Capacity	Energy Density
1	50.4	44.4	7C	16320	16310
2	50.4	44.4	7C	16310	16300
3	50.4	44.4	7C	16300	16290
4	50.4	44.4	7C	16290	16280
5	50.4	44.4	7C	16280	16270
6	50.4	44.4	7C	16270	16260
7	50.4	44.4	7C	16260	16250
8	50.4	44.4	7C	16250	16240
9	50.4	44.4	7C	16240	16230
10	50.4	44.4	7C	16230	16220
11	50.4	44.4	7C	16220	16210
12	50.4	44.4	7C	16210	16200
13	50.4	44.4	7C	16200	16190
14	50.4	44.4	7C	16190	16180
15	50.4	44.4	7C	16180	16170
16	50.4	44.4	7C	16170	16160
17	50.4	44.4	7C	16160	16150
18	50.4	44.4	7C	16150	16140
19	50.4	44.4	7C	16140	16130
20	50.4	44.4	7C	16130	16120
21	50.4	44.4	7C	16120	16110
22	50.4	44.4	7C	16110	16100
23	50.4	44.4	7C	16100	16090
24	50.4	44.4	7C	16090	16080
25	50.4	44.4	7C	16080	16070
26	50.4	44.4	7C	16070	16060
27	50.4	44.4	7C	16060	16050
28	50.4	44.4	7C	16050	16040
29	50.4	44.4	7C	16040	16030
30	50.4	44.4	7C	16030	16020

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing





TC-5696

Test Report No: NEXT-20220713-04

IS 16047:2012/IEC61960:2011

Page 4 of 8

ULR: TC569622100001686F

Date: 08/11/2022

Group: Cells & Batteries

Discipline: Electrical

31	50.4	44.4	7C	16020	16010
32	50.4	44.4	7C	16010	16000
33	50.4	44.4	7C	16000	15990
34	50.4	44.4	7C	15990	15980
35	50.4	44.4	7C	15980	15970
36	50.4	44.4	7C	15970	15960
37	50.4	44.4	7C	15960	15950
38	50.4	44.4	7C	15950	15940
39	50.4	44.4	7C	15940	15930
40	50.4	44.4	7C	15930	15920
41	50.4	44.4	7C	15920	15910
42	50.4	44.4	7C	15910	15900
43	50.4	44.4	7C	15900	15890
44	50.4	44.4	7C	15890	15880
45	50.4	44.4	7C	15880	15870
46	50.4	44.4	7C	15870	15860
47	50.4	44.4	7C	15860	15850
48	50.4	44.4	7C	15850	15840
49	50.4	44.4	7C	15840	15830
50	50.3	44.4	7C	15830	15820
51	50.3	44.4	7C	15820	15810
52	50.3	44.4	7C	15810	15800
53	50.3	44.4	7C	15800	15790
54	50.3	44.4	7C	15790	15780
55	50.3	44.4	7C	15780	15770
56	50.3	44.4	7C	15770	15760
57	50.3	44.4	7C	15760	15750
58	50.3	44.4	7C	15750	15740
59	50.3	44.4	7C	15740	15730
60	50.3	44.4	7C	15730	15720
61	50.3	44.4	7C	15720	15710
62	50.3	44.4	7C	15710	15700
63	50.3	44.4	7C	15700	15690
64	50.3	44.4	7C	15690	15680
65	50.3	44.4	7C	15680	15670
66	50.3	44.4	7C	15670	15660
67	50.3	44.4	7C	15660	15650
68	50.3	44.4	7C	15650	15640
69	50.3	44.4	7C	15640	15630
70	50.3	44.4	7C	15630	15620
71	50.3	44.4	7C	15620	15610
72	50.3	44.4	7C	15610	15600
73	50.3	44.4	7C	15600	15590
74	50.3	44.4	7C	15590	15580
75	50.3	44.4	7C	15580	15570
76	50.3	44.4	7C	15570	15560

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing





TC-5696

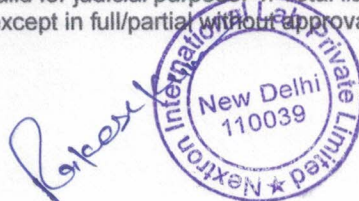
Test Report No: NEXT-20220713-04
 ULR: TC569622100001686F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 5 of 8
 Date: 08/11/2022
 Discipline: Electrical

77	50.3	44.4	7C	15560	15550
78	50.3	44.4	7C	15550	15540
79	50.3	44.4	7C	15540	15530
80	50.3	44.4	7C	15530	15520
81	50.3	44.4	7C	15520	15510
82	50.3	44.4	7C	15510	15500
83	50.3	44.4	7C	15500	15490
84	50.3	44.4	7C	15490	15480
85	50.3	44.4	7C	15480	15470
86	50.3	44.4	7C	15470	15460
87	50.3	44.4	7C	15460	15450
88	50.3	44.4	7C	15450	15440
89	50.3	44.4	7C	15440	15430
90	50.3	44.4	7C	15430	15420
91	50.3	44.4	7C	15420	15410
92	50.3	44.4	7C	15410	15400
93	50.3	44.4	7C	15400	15390
94	50.3	44.4	7C	15390	15380
95	50.3	44.4	7C	15380	15370
96	50.3	44.4	7C	15370	15360
97	50.3	44.4	7C	15360	15350
98	50.3	44.4	7C	15350	15340
99	50.3	44.4	7C	15340	15330
100	50.2	44.4	7C	15330	15320
101	50.2	44.4	7C	15320	15310
102	50.2	44.4	7C	15310	15300
103	50.2	44.4	7C	15300	15290
104	50.2	44.4	7C	15290	15280
105	50.2	44.4	7C	15280	15270
106	50.2	44.4	7C	15270	15260
107	50.2	44.4	7C	15260	15250
108	50.2	44.4	7C	15250	15240
109	50.2	44.4	7C	15240	15230
110	50.2	44.4	7C	15230	15220
111	50.2	44.4	7C	15220	15210
112	50.2	44.4	7C	15210	15200
113	50.2	44.4	7C	15200	15190
114	50.2	44.4	7C	15190	15180
115	50.2	44.4	7C	15180	15170
116	50.2	44.4	7C	15170	15160
117	50.2	44.4	7C	15160	15150
118	50.2	44.4	7C	15150	15140
119	50.2	44.4	7C	15140	15130
120	50.2	44.4	7C	15130	15120
121	50.2	44.4	7C	15120	15110
122	50.2	44.4	7C	15110	15100

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing





TC-5696

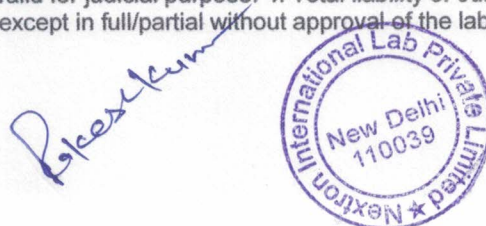
Test Report No: NEXT-20220713-04
 ULR: TC569622100001686F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 6 of 8
 Date: 08/11/2022
 Discipline: Electrical

123	50.2	44.4	7C	15100	15090
124	50.2	44.4	7C	15090	15080
125	50.2	44.4	7C	15080	15070
126	50.2	44.4	7C	15070	15060
127	50.2	44.4	7C	15060	15050
128	50.2	44.4	7C	15050	15040
129	50.2	44.4	7C	15040	15030
130	50.2	44.4	7C	15030	15020
131	50.2	44.4	7C	15020	15010
132	50.2	44.4	7C	15010	15000
133	50.2	44.4	7C	15000	14990
134	50.2	44.4	7C	14990	14980
135	50.2	44.4	7C	14980	14970
136	50.2	44.4	7C	14970	14960
137	50.2	44.4	7C	14960	14950
138	50.2	44.4	7C	14950	14940
139	50.2	44.4	7C	14940	14930
140	50.2	44.4	7C	14930	14920
141	50.2	44.4	7C	14920	14910
142	50.2	44.4	7C	14910	14900
143	50.2	44.4	7C	14900	14890
144	50.2	44.4	7C	14890	14880
145	50.2	44.4	7C	14880	14870
146	50.2	44.4	7C	14870	14860
147	50.2	44.4	7C	14860	14850
148	50.2	44.4	7C	14850	14840
149	50.2	44.4	7C	14840	14830
150	50.2	44.4	7C	14830	14820
151	50.2	44.4	7C	14820	14810
152	50.2	44.4	7C	14810	14800
153	50.2	44.4	7C	14800	14790
154	50.2	44.4	7C	14790	14780
155	50.2	44.4	7C	14780	14770
156	50.2	44.4	7C	14770	14760
157	50.2	44.4	7C	14760	14750
158	50.1	44.4	7C	14750	14740
159	50.1	44.4	7C	14740	14730
160	50.1	44.4	7C	14730	14720
161	50.1	44.4	7C	14720	14710
162	50.1	44.4	7C	14710	14700
163	50.1	44.4	7C	14700	14690
164	50.1	44.4	7C	14690	14680
165	50.1	44.4	7C	14680	14670
166	50.1	44.4	7C	14670	14660
167	50.1	44.4	7C	14660	14650
168	50.1	44.4	7C	14650	14640

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing





TC-5696

Test Report No: NEXT-20220713-04
 ULR: TC569622100001686F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 7 of 8
 Date: 08/11/2022
 Discipline: Electrical

169	50.1	44.4	7C	14640	14630
170	50.1	44.4	7C	14630	14620
171	50.1	44.4	7C	14620	14610
172	50.1	44.4	7C	14610	14600
173	50.1	44.4	7C	14600	14590
174	50.1	44.4	7C	14590	14580
175	50.1	44.4	7C	14580	14570
176	50.1	44.4	7C	14570	14560
177	50.1	44.4	7C	14560	14550
178	50.1	44.4	7C	14550	14540
179	50.1	44.4	7C	14540	14530
180	50.1	44.4	7C	14530	14520
181	50.1	44.4	7C	14520	14510
182	50.1	44.4	7C	14510	14500
183	50.1	44.4	7C	14500	14490
184	50.1	44.4	7C	14490	14480
185	50.1	44.4	7C	14480	14470
186	50.1	44.4	7C	14470	14460
187	50.1	44.4	7C	14460	14450
188	50.1	44.4	7C	14450	14440
189	50.1	44.4	7C	14440	14430
190	50.1	44.4	7C	14430	14420
191	50.1	44.4	7C	14420	14410
192	50.1	44.4	7C	14410	14400
193	50.1	44.4	7C	14400	14390
194	50.1	44.4	7C	14390	14380
195	50.1	44.4	7C	14380	14370
196	50.1	44.4	7C	14370	14360
197	50.1	44.4	7C	14360	14350
198	50.1	44.4	7C	14350	14340
199	50.1	44.4	7C	14340	14330
200	50.1	44.4	7C	14330	14320
201	50.1	44.4	7C	14320	14310
202	50.1	44.4	7C	14310	14300
203	50.1	44.4	7C	14300	14290
204	50.1	44.4	7C	14290	14280
205	50.1	44.4	7C	14280	14270
206	50.1	44.4	7C	14270	14260
207	50.1	44.4	7C	14260	14250
208	50.1	44.4	7C	14250	14240
209	50.1	44.4	7C	14240	14230
210	50	44.4	7C	14230	14220
211	50	44.4	7C	14220	14210
212	50	44.4	7C	14210	14200
213	50	44.4	7C	14200	14190
214	50	44.4	7C	14190	14180

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing

Rakesh Kumar





TC-5696

Test Report No: NEXT-20220713-04
 ULR: TC569622100001686F
 Group: Cells & Batteries

IS 16047:2012/IEC61960:2011

Page 8 of 8

Date: 08/11/2022

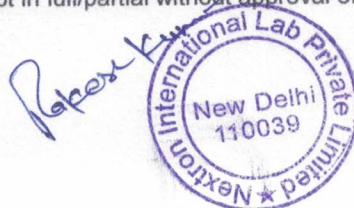
Discipline: Electrical

215	50	44.4	7C	14180	14170
216	50	44.4	7C	14170	14160
217	50	44.4	7C	14160	14150
218	50	44.4	7C	14150	14140
219	50	44.4	7C	14140	14130
220	50	44.4	7C	14130	14120
221	50	44.4	7C	14120	14110
222	50	44.4	7C	14110	14100
223	50	44.4	7C	14100	13390
224	50	44.4	7C	13390	13380
225	50	44.4	7C	13380	13370
226	50	44.4	7C	13370	13360
227	50	44.4	7C	13360	13350
228	50	44.4	7C	13350	13340
229	50	44.4	7C	13340	13330
230	50	44.4	7C	13330	13320
231	50	44.4	7C	13320	13310
232	50	44.4	7C	13310	13300
233	50	44.4	7C	13300	13290
234	50	44.4	7C	13290	13280
235	50	44.4	7C	13280	13270
236	50	44.4	7C	13270	13260
237	50	44.4	7C	13260	13250
238	50	44.4	7C	13250	13240
239	50	44.4	7C	13240	13230
240	50	44.4	7C	13230	13220
241	50	44.4	7C	13220	13210
242	50	44.4	7C	13210	13200
243	50	44.4	7C	13200	13190
244	50	44.4	7C	13190	13180
245	50	44.4	7C	13180	13170
246	50	44.4	7C	13170	13160
247	50	44.4	7C	13160	13150
248	50	44.4	7C	13150	13140
249	50	44.4	7C	13140	13130
250	50	44.4	7C	13130	13120

Charge Discharge Cycle (Battery Life Cycle Test)		
Test Parameters	Charge Capacity	Discharge Capacity
Initial Capacity	16.32Ah	16.31Ah
Capacity After "Declared Value by manufacturer" Cycles	13.13Ah	13.12Ah
Capacity Determination	16Ah	16Ah
Any other parameters	N/A	N/A
Remarks If any	Nil	Nil

END OF TEST REPORT

1. Test Report are valid only for the sample tested in our laboratory. 2. Samples tested are Non-returnable until specified at a time of sample submission. 3. This report will not be valid for judicial purpose. 4. Total liability of our test lab is limited to invoiced amount 5. This report is not to be reproduced except in full/partial without approval of the laboratory in writing





Test Report

Report No.: ITI/EMC/TR/FC/012	ULR No.:TC979722000000007F	Date:26/03/2022
Job Card Number	059	
Test Requested by (Name of Organisation)	Garuda Aerospace	
Address of the Organisation	Garuda Aerospace Pvt. Ltd 24, 46, KB Dasan Rd, Sektammal Colony, Lubdhi Colony, Alwarpet, Chennai, Tamil Nadu 600018	
Description/Unambiguous Identification of the item:		
Name of the device	Multicopter Drone Controller	
Manufactured by	Garuda Aerospace	
Model no.		
No. of items submitted	1	
Serial no.		
Applicable standard/ test specification	(1) IEC 61000-4-2:2008 (2) IEC 61000-4-3 Ed. 4.0:2020-09,	
Testing Laboratory information:		
Testing Laboratory Name	EMC Lab, ITI Limited	
Address:	F53, TELECOM TEST LABS, ITI LIMITED, DOORAVANINAGAR, BENGALURU, BENGALURU URBAN, KARNATAKA, INDIA, 560016	

This report refers only to the DUT tested in EMC Lab at ITI Limited, Bangalore. The report cannot be duplicated without the approval from the Head of EMC Lab, ITI Limited, Bangalore.

Testing	
Date of submission of test samples	07/02/2022
Date of start of tests	10/02/2022





EMC LAB
ITI LIMITED
(A Government of India Undertaking)
Dooravaninagar, Bangalore -560 016
Phone No : (080) 28503695
Email: emiemo_bgp@itild.co.in



TC-8797

Date of end of tests	10/02/2022
Date of issue of test report	26/03/2022
Test Category	Full Compliance
Laboratory Environment condition	
Declaration of Conformity:	Declaration of conformity of the results is based as per the standard limits.
Authorized Signatory	
Prepared by : Deepak Kumar ASSISTANT EXECUTIVE ENGINEER	<i>Deepak Kumar</i> 26/03/2022
Reviewed by : Prateek Kumar ASSISTANT EXECUTIVE ENGINEER	<i>Prateek Kumar</i> 26/3/22
Authorized & Issued by : Narashimappa N S TECHNICAL MANAGER	<i>Narashimappa N S</i> 26/03/22



Contents

<u>CONTENTS</u>	Page
1. TEST SUMMARY	3
1.1 MEASUREMENT UNCERTAINTY	4
2. TEST FACILITY	6
2.1 DEVIATION FROM STANDARD	7
2.2 ABNORMALITIES FROM STANDARD CONDITIONS	7
3. GENERAL INFORMATION	8
3.1 GENERAL DESCRIPTION OF EUT	8
4. EQUIPMENT LIST FOR ALL TEST ITEMS	9
5. IMMUNITY TEST RESULTS	10
5.1 ELECTROSTATIC DISCHARGE IMMUNITY TEST	10
5.1.1 E.U.T. Operation	10
5.1.2 Test specification	10
5.1.3 Measurement Data	11
5.1.4 Test Setup Photograph	12
5.2 RF FIELD STRENGTH IMMUNITY TEST	14
5.2.1 E.U.T. Operation	14
5.2.2 Test specification	14
5.2.3 Measurement Data	15
5.2.4 Test Setup Photograph	15





TC-8787

1. Test Summary

Name of the Test	Standard/ Method	Limits	Performance Criteria Required	Results
Radiated Susceptibility	IEC 61000-4-3 Ed. 4.0:2020-09	80MHz-6 GHz, 3V/m (Level 2)	A	PASS (Meets the performance criteria A)
Electrostatic Discharge test	IEC 61000-4-2:2008	Air Discharge: ± 4 kV Contact Discharge: ± 2 kV	A	PASS (Meets the performance criteria A)





1.1 Measurement Uncertainty

The report uncertainty of measurement $y \pm U$, where expanded uncertainty U is based on a standard uncertainty Multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95%.

No.	Item	Parameter	U, Value
1	Electro Static Discharge	2kV	179 ps (Rise time)





TC-9797

2. Test Facility

The test facility is recognized, certified or accredited by the following organizations:

ILAC / NABL Accreditation No.: TC-9797

EMC Lab, ITI Limited has been registered by National Accreditation Board for Testing and Calibration Laboratories (NABL).

TEC Designation No.: TEC/MRA/CAB/IND-D/79

EMC Lab, ITI Limited has been registered by Telecommunication Engineering (TEC) Center.

2.1 Deviation from standard

None

2.2 Abnormalities from standard conditions

None



3. General Information

3.1 General Description of EUT

Manufacturer:	Garuda Aerospace
Manufacturer Address:	Garuda Aerospace Pvt. Ltd 24, 46, KB Dasan Rd, Seetammal Colony, Lubdhi Colony, Alwarpet, Chennai, Tamil Nadu 600018
EUT Name:	Agriculture Drone
Model No:	NA
Serial Number:	NA
Associate model:	NA
Brand Name:	NA
H/W No.:	NA
S/W No.:	NA
Marking Level:	NA
Power Supply Range:	Nominal Input: 3 V DC
Battery:	YES



4. Equipment List for All Test Items

ESD Test Equipment						
No	Test Equipment	Manufacturer	Model No	Serial No	Cal. Date	Cal. Due Date
1	ESD Gun	EMC Partner (ESD3000)	103605	2028	19.08.2020	19.08.2022

RS Test Equipment						
No	Test Equipment	Manufacturer	Model No	Serial No	Cal. Date	Cal. Due Date
1	Signal Generator	Keysight	N5171B-506	MY57280520	17.03.2021	17.03.2024
2	RF Amplifier (80 MHz to 1 GHz)	ETSL	8000-004	00163068	NA	NA
3	RF Amplifier (1 MHz to 6 GHz)	ETSL	8100-008	00163576	NA	NA
4	Stacked Log Periodic Antenna	Schwarz beck	STLP 9128E	9128E-73033	08.06.2018	08.06.2022
5	Horn Antenna	Schwarz beck	BBHA9120E	789	08.06.2018	08.06.2022



5. Immunity Test Results

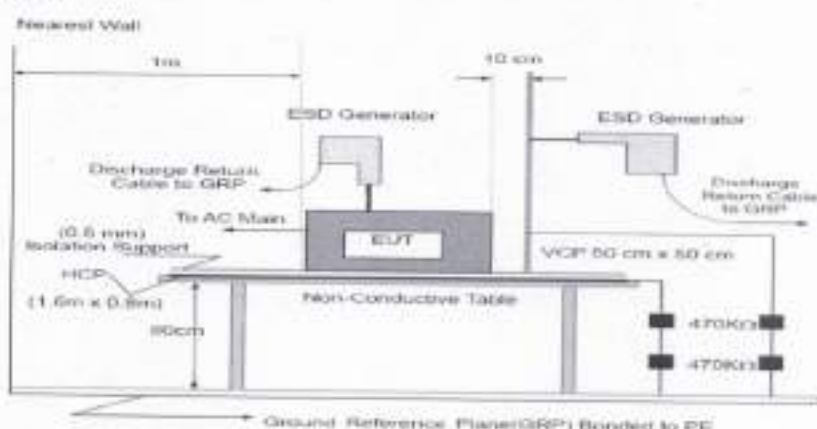
5.1 Electrostatic discharge immunity test

Acceptable Performance Criterion:	A
Applicable Standard:	IEC 61000-4-2:2008
Discharge Impedance:	330 Ω /150 pF
Discharge Voltage:	Air Discharge: ± 4 kV (Level 2)
	Contact Discharge: ± 2 kV (Level 1)
	VCP, HCP: YES
Polarity:	Positive & Negative
Minimum discharge Interval:	1 second

5.1.1 E.U.T. Operation

Temperature:	25.3°C	Humidity:	54% RH	Atmospheric Pressure:	96.52	Kpa
Test Mode:				NA		

5.1.2 Test specification



EUT was operated in the mode as mentioned above. Both contact and air discharge was executed. Contact discharge to the conductive surfaces; air discharge at insulating surfaces. Each test point shall be subjected to 05 discharges at least (For each voltage and polarity).

5.1.3 Measurement Data

Test Record

Electrostatic Discharge Test Results																		
M/N:											Test Result: ☒ Pass ☐ Fail							
Test Voltage:	3 V DC										Test date:09-02-2022							
Discharge times	Contact discharge: minimum 05 times (+/- respectively) at each point, Air discharge: minimum 05 times (+/- respectively) at each point.																	
Discharge Mode	Air Discharge								Contact Discharge								Required Performance Criterion	Result
Test level (kV)	2		4		6		8		2		4		6		8			
Test Location	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-		
HCP	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-	A	Pass
VCP	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-		Pass
screws	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-		Pass
Enclosure	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-		Pass
LED	A	A	A	A	-	-	-	-	-	-	-	-	-	-	-	-		Pass
Plastic part	A	A	A	A	-	-	-	-	-	-	-	-	-	-	-	-		Pass
Note: Horizontal Coupling Plane (HCP) and Vertical Coupling plane (VCP). "Cx" means Contact Point, x=1~N, "Ax" means Air Point, x=1~N.																		



5.1.4 Test Setup Photograph



Fig 1 Test setup Air Discharge





Fig 2: Test Setup Contact Discharge



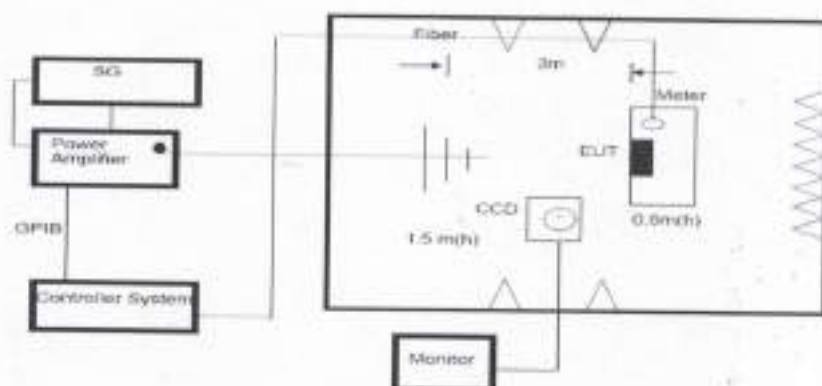
5.2 RF field strength immunity test

Acceptable Performance Criterion:	A
Applicable Standard:	IEC 61000-4-3:2020
Frequency Range & Test Level	80MHz~6 GHz, 3V/m (Level 2)
Test Distance	3 m
Polarity:	Horizontal & Vertical with all four orientation

5.2.1 E.U.T. Operation

Temperature:	24°C	Humidity:	53% RH	Atmospheric Pressure:	97.4	Kpa
Test Mode:	NA					

5.2.2 Test specification



Test was executed in a fully Anechoic chamber. An antenna was used to transmit interference signal. EUT was placed upon a wooden table above the reference ground 0.8m, and was positioned so that the four sides of the EUT shall be exposed to the electromagnetic field in a sequence. In each position the performance of the EUT was investigated. A camera was used to monitor the loss of function or degradation of performance of the EUT.

5.2.3 Measurement Data

Test Record

Test Voltage:	3 V DC		
Test Port	Enclosure		
Operating Mode	NA		
Test Level	3 V/m(r.m.s) (Modulated)		A
Frequency Range(MHz)	Antenna polarity	Modulation	Result
80 MHz~1GHz	H & V	1KHz, 80% AM & 3 V/m	Pass
1 GHz ~ 6 GHz	H & V	1KHz, 80% AM & 3V/m	Pass

5.2.4 Test Setup Photograph



Fig 3: Test setup for Radiated Immunity Test-H



Fig 4: Test setup for Radiated Immunity Test with Log periodic Antenna-H

*****END OF REPORT*****





Test Report

Report No.: ITI/EMC/TR/FC/011	ULR No.:TC979722000000006F	Date: 26/03/2022
Job Card Number	059	
Test Requested by (Name of Organisation)	Garuda Aerospace	
Address of the Organisation	Garuda Aerospace Pvt. Ltd 24, 46, KB Dasan Rd, Seetammal Colony, Lubdhi Colony, Alwarpet, Chennai, Tamil Nadu 600018	
Description/Unambiguous Identification of the item:		
Name of the device	Multicopter Drone	
Manufactured by	Garuda Aerospace	
Model no.	GA-AD	
No. of items submitted	1	
Serial no.	003	
Applicable standard/ test specification	(1) IEC 61000-4-2:2008 (2) IEC 61000-4-3 Ed. 4.0:2020-09,	
Testing Laboratory information:		
Testing Laboratory Name	EMC Lab, ITI Limited	
Address:	F53, TELECOM TEST LABS, ITI LIMITED, DOORAVANINAGAR, BENGALURU, BENGALURU URBAN, KARNATAKA, INDIA, 560016	
This report refers only to the DUT tested in EMC Lab at ITI Limited, Bangalore. The report cannot be duplicated without the approval from the Head of EMC Lab, ITI Limited, Bangalore.		
Testing		
Date of submission of test samples	07/02/2022	
Date of start of tests	10/02/2022	

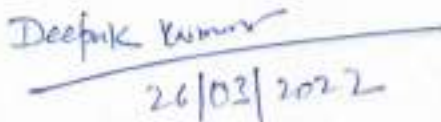




EMC LAB
ITI LIMITED
(A Government of India Undertaking)
Dooravaninagar, Bangalore -560 016
Phone No : (080) 28503695
Email: emienc_bgp@itilttd.co.in



TC-8787

Date of end of tests	10/02/2022
Date of issue of test report	26/03/2022
Test Category	Full Compliance
Laboratory Environment condition	
Declaration of Conformity:	Declaration of conformity of the results is based as per the standard limits.
Authorized Signatory	
Prepared by : Deepak Kumar ASSISTANT EXECUTIVE ENGINEER	 26/03/2022
Reviewed by : Prateek Kumar ASSISTANT EXECUTIVE ENGINEER	 26/3/22
Authorized & Issued by : Narashimappa N S TECHNICAL MANAGER	



Contents

<u>CONTENTS</u>	Page
1. TEST SUMMARY	3
1.1 MEASUREMENT UNCERTAINTY	4
2. TEST FACILITY	6
2.1 DEVIATION FROM STANDARD	7
2.2 ABNORMALITIES FROM STANDARD CONDITIONS	7
3. GENERAL INFORMATION	8
3.1 GENERAL DESCRIPTION OF EUT	8
3.2 EUT DESCRIPTION:-	8
5 IMMUNITY TEST RESULTS	10
6.1 ELECTROSTATIC DISCHARGE IMMUNITY TEST	10
6.1.1 E.U.T. Operation	10
6.1.2 Test specification	10
6.1.3 Measurement Data	11
6.1.4 Test Setup Photograph	12
6.2 RF FIELD STRENGTH IMMUNITY TEST	14
6.2.1 E.U.T. Operation	14
6.2.2 Test specification	14
6.2.3 Measurement Data	15
6.2.4 Test Setup Photograph	15





EMC LAB
ITI LIMITED
(A Government of India Undertaking)
Dooravaninagar, Bangalore -560 016
Phone No : (080) 28503695
Email: emienc_bgp@itiltd.co.in



TC-9797

1. Test Summary

Name of the Test	Standard/ Method	Limits	Performance Criteria Required	Results
Radiated Susceptibility	IEC 61000-4-3 Ed. 4.0:2020-09	80MHz-6 GHz,3V/m, (Level 2)	A	PASS (Meets the performance criteria A)
Electrostatic Discharge test	IEC 61000-4-2:2008	Air Discharge: $\pm 4\text{kV}$ Contact Discharge: $\pm 2\text{kV}$	A	PASS (Meets the performance criteria A)





1.1 Measurement Uncertainty

The report uncertainty of measurement $y \pm U$, where expanded uncertainty U is based on a standard uncertainty Multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95%.

No.	Item	Parameter	U, Value
1	Electro Static Discharge	2kV	179 ps (Rise time)





2. Test Facility

The test facility is recognized, certified or accredited by the following organizations:

ILAC / NABL Accreditation No.: TC-9797

EMC Lab, ITI Limited has been registered by National Accreditation Board for Testing and Calibration Laboratories (NABL).

TEC Designation No.: TEC/MRA/CAB/IND-D/79

EMC Lab, ITI Limited has been registered by Telecommunication Engineering (TEC) Center.

2.1 Deviation from standard

None

2.2 Abnormalities from standard conditions

None



3. General Information

3.1 General Description of EUT

Manufacturer:	Garuda Aerospace
Manufacturer Address:	Garuda Aerospace Pvt. Ltd 24, 46, KB Dasan Rd, Seetammal Colony, Lubdhi Colony, Alwarpet, Chennai, Tamil Nadu 600018
EUT Name:	Multicopter Drone
Model No:	GA-AD
Serial Number:	003
Associate model:	NA
Brand Name:	Garuda Aerospace
H/W No.:	NA
S/W No.:	NA
Marking Level:	NA
Power Supply Range:	Nominal Input: 3 V DC
Battery:	YES

3.2 EUT Description:-

Garuda drones is equipped with a spraying system which is capable of spraying liquids at variable conditions. The nozzles are of range 5-20microns which is helpful in efficient spraying over the crops.



4. Equipment List for All Test Items

ESD Test Equipment						
No	Test Equipment	Manufacturer	Model No	Serial No	Cal. Date	Cal. Due Date
1	ESD Gun	EMC Partner (ESD3000)	103605	2028	19.08.2020	19.08.2023

RS Test Equipment						
No	Test Equipment	Manufacturer	Model No	Serial No	Cal. Date	Cal. Due Date
1	Signal Generator	Keysight	N5171B-506	MY57280520	17.03.2021	17.03.2024
2	RF Amplifier (80 MHz to 1 GHz)	ETSL	8000-004	00163068	NA	NA
3	RF Amplifier (1 MHz to 6 GHz)	ETSL	8100-008	00163576	NA	NA
4	Stacked Log Periodic Antenna	Schwarz beck	STLP 9128E	9128E-73033	08.06.2018	08.06.2022
5	Horn Antenna	Schwarz beck	BBHA9120E	789	08.06.2018	08.06.2022



5. Immunity Test Results

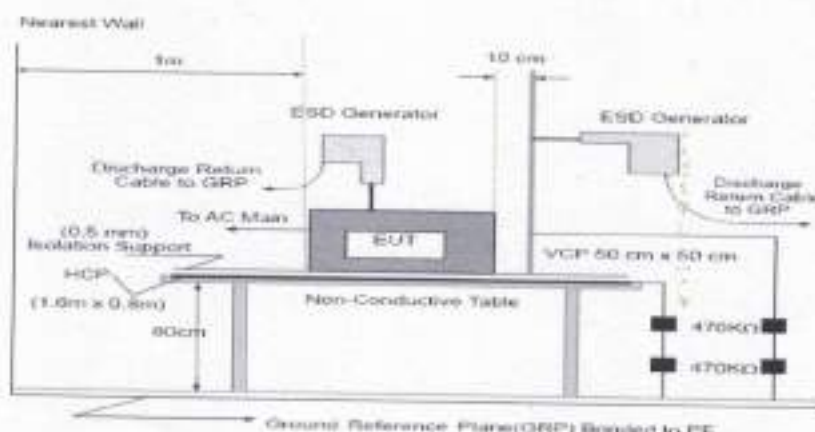
5.1 Electrostatic discharge immunity test

Acceptable Performance Criterion:	B
Applicable Standard:	IEC 61000-4-2:2008
Discharge Impedance:	330 Ω /150 pF
Discharge Voltage:	Air Discharge: ± 4 kV (Level 2)
	Contact Discharge: ± 2 kV (Level 1)
	VCP, HCP: N/A
Polarity:	Positive & Negative
Minimum discharge interval:	1 second

5.1.1 E.U.T. Operation

Temperature:	25.3°C	Humidity:	54% RH	Atmospheric Pressure:	96.52	Kpa
Test Mode:				NA		

5.1.2 Test specification



EUT was operated in the mode as mentioned above. Both contact and air discharge was executed. Contact discharge to the conductive surfaces; air discharge at insulating surfaces. Each test point shall be subjected to 05 discharges at least (For each voltage and polarity).



5.1.3 Measurement Data

Test Record

Electrostatic Discharge Test Results																			
M/N:											Test Result: ☒ Pass ☐ Fail								
Test Voltage:	3 V DC										Test date:09-02-2022								
Discharge times	Contact discharge: minimum 05 times (+/- respectively) at each point, Air discharge: minimum 05 times (+/- respectively) at each point.																		
Discharge Mode	Air Discharge								Contact Discharge								Required Performance Criterion	Result	
Test level (kV)	2		4		6		8		2		4		6		8				
Test Location	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-			
HCP	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-	-	A	Pass
VCP	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-	-		Pass
screws	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-	-		Pass
Enclosure	-	-	-	-	-	-	-	-	A	A	-	-	-	-	-	-	-		Pass
LED	A	A	A	A	-	-	-	-	-	-	-	-	-	-	-	-	-		Pass
Plastic part	A	A	A	A	-	-	-	-	-	-	-	-	-	-	-	-	-		Pass

Note: Horizontal Coupling Plane (HCP) and Vertical Coupling plane (VCP).
 "Cx" means Contact Point, x=1~N, "Ax" means Air Point, x=1~N.



5.1.4 Test Setup Photograph



Fig 1: Test setup For Air Discharge



Fig 2: Test Setup For Contact Discharge

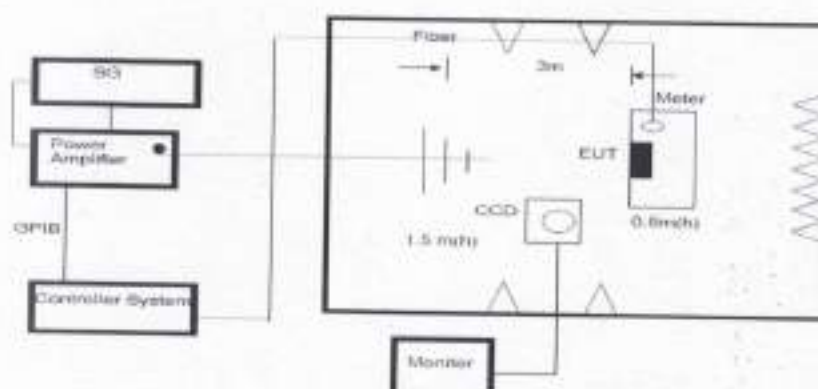
5.2 RF field strength immunity test

Acceptable Performance Criterion:	A
Applicable Standard:	IEC 61000-4-3:2020
Frequency Range & Test Level	80MHz~6 GHz, 3V/m (Level 2)
Test Distance	3 m
Polarity:	Horizontal & Vertical with all four orientation

5.2.1 E.U.T. Operation

Temperature:	24°C	Humidity:	53% RH	Atmospheric Pressure:	97.4	Kpa
Test Mode:	NA					

5.2.2 Test specification



Test was executed in a fully Anechoic chamber. An antenna was used to transmit interference signal. EUT was placed upon a wooden table above the reference ground 0.8m, and was positioned so that the four sides of the EUT shall be exposed to the electromagnetic field in a sequence. In each position the performance of the EUT was investigated. A camera was used to monitor the loss of function or degradation of performance of the EUT.

5.2.3 Measurement Data

Test Record

Test Voltage:	3 V DC		
Test Port	Enclosure		
Operating Mode	NA		
Test Level	3 V/m(r.m.s) (Modulated)		A
Frequency Range(MHz)	Antenna polarity	Modulation	Result
80 MHz~1GHz	H &V	1KHz, 80% AM & 3 V/m	Pass
1 GHz ~ 6 GHz	H &V	1KHz, 80% AM & 3V/m	Pass

5.2.4 Test Setup Photograph



Fig 3: Test setup for Radiated Immunity Test-H Polarization



Fig 4: Test setup for Radiated Immunity Test with Log periodic Antenna-H Polarization

*****END OF REPORT*****



	TEST REPORT		
	TEST CODE: HIGH TEMPERATURE OPERATION		
	Test Report No: 009/2203/HTO/GARUDA ULR: TC507822000000150F		

HIGH TEMPERATURE OPERATION TEST REPORT

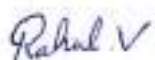
Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation.

NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.

	TEST REPORT		
	TEST CODE: HIGH TEMPERATURE STORAGE		
	Test Report No: 011/2203/HTS/GARUDA ULR: TC507822000000153F		

HIGH TEMPERATURE STORAGE TEST REPORT

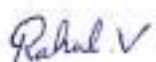
Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation.

NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.

	TEST REPORT		
	TEST CODE: LOW TEMPERATURE OPERATION		
	Test Report No: 012/2203/LTO/GARUDA ULR: TC507822000000154F		

LOW TEMPERATURE OPERATION TEST REPORT

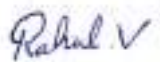
Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation.

NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.

	TEST REPORT		
	TEST CODE: LOW TEMPERATURE STORAGE		
	Test Report No: 014/2203/LTS/GARUDA ULR: TC507822000000156F		

LOW TEMPERATURE STORAGE TEST REPORT

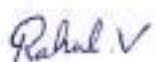
Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation.

NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.

	TEST REPORT	
	TEST CODE: DAMP HEAT	
	Test Report No: 015/2203/DH/GARUDA ULR: TC507822000000157F	

DAMP HEAT TEST REPORT

Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation. **NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.**

	TEST REPORT		
	TEST CODE: THERMAL CYCLING		
	Test Report No: 017/2203/TC/GARUDA ULR: TC507822000000159F		

THERMAL CYCLING TEST REPORT

Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 07th Jun 2022

Prepared By,



Rahul V Gowda
Testing Engineer

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 07-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation. **NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.**

	TEST REPORT		
	TEST CODE: MECHANICAL SHOCK		
	Test Report No: 060/2206/MS/GARUDA ULR: TC507822000000478F		

MECHANICAL SHOCK TEST REPORT

Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 18th Jun 2022

Prepared By,



Rohit.K
Lab Technician

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 18-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation. **NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.**

	TEST REPORT		
	TEST CODE: SINUSOIDAL VIBRATION		
	Test Report No: 061/2206/VIB/GARUDA ULR: TC50782200000049F		

SINUSOIDAL VIBRATION TEST REPORT

Submitted to



Garuda Aerospace Pvt. Ltd
24, 46, KB Dasan Rd, Seetammal Colony,
Lubdhi Colony, Alwarpet, Chennai,
Tamil Nadu 600018

Original Issue Date: 18th Jun 2022

Prepared By,



Rohit.K
Lab Technician

Approved By,



Anil Kumar Ammina
Quality Manager

Date: 18-06-2022

Disclaimer: In making any use of this test report you are expressly agreeing to the disclaimers and notices below: BE Analytic Testing procedures were performed to the best of our ability. All data and test results presented in your report document the equipment's characteristics which were analyzed and/or calculated at that specific point in time. All test data and results presented are pertaining only to the DUT/SAMPLE, which was received for the test with identity mentioned in the below report. This particular test does not require any statement of conformity or opinion/interpretation. **NOTE: This report shall not be reproduced except in full without approval of the laboratory can provide assurance that parts of a report are not taken out of context.**